

EXP : 11

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import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Sample dataset
data = {
    'Income': [25000, 30000, 45000, 50000, 60000, 75000, 80000, 90000,
               28000, 40000, 55000, 70000],
    'Debt': [20000, 18000, 15000, 12000, 10000, 8000, 5000, 4000,
            22000, 16000, 11000, 7000],
    'Payment_History': [1, 1, 2, 2, 2, 3, 3, 3, 1, 2, 2, 3],
    'Credit_Utilization': [90, 80, 70, 60, 50, 40, 30, 20, 85, 65, 45, 35],
    'Credit_Score': [
        'Poor', 'Poor', 'Average', 'Average',
        'Average', 'Good', 'Good', 'Good',
        'Poor', 'Average', 'Average', 'Good'
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

# Input features and target
X = df[['Income', 'Debt', 'Payment_History', 'Credit_Utilization']]
y = df['Credit_Score']

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
```

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X, y, test_size=0.25, random_state=42
)

# Create Decision Tree model
model = DecisionTreeClassifier(random_state=42)

# Train the model
model.fit(X_train, y_train)

# Predict credit scores
y_pred = model.predict(X_test)

# Display predictions
print("Actual Credit Scores:")
print(y_test.values)

print("\nPredicted Credit Scores:")
print(y_pred)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")

# Display confusion matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
... Actual Credit Scores:
['Average' 'Average' 'Poor']

Predicted Credit Scores:
['Average' 'Average' 'Poor']

Accuracy: 100.0 %

Confusion Matrix:
[[2 0]
 [0 1]]
```